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FET-UNet: Merging CNN and transformer architectures for superior breast ultrasound image segmentation

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ABSTRACT

Purpose: Breast cancer remains a significant cause of mortality among women globally, highlighting the critical need for accurate diagnosis. Although Convolutional Neural Networks (CNNs) have shown effectiveness in segmenting breast ultrasound images, they often face challenges in capturing long-range dependencies, particularly for lesions with similar intensity distributions, irregular shapes, and blurred boundaries. To overcome these limitations, we introduce FET-UNet, a novel hybrid framework that integrates CNNs and Swin Transformers within a UNet-like architecture.
Methods: FET-UNet features parallel branches for feature extraction: one utilizes ResNet34 blocks, and the other employs Swin Transformer blocks. These branches are fused using an advanced feature aggregation module (AFAM), enabling the network to effectively combine local details and global context. Additionally, we include a multi-scale upsampling mechanism in the decoder to ensure precise segmentation outputs. This design enhances the capture of both local details and long-range dependencies.
Results: Extensive evaluations on the BUSI, UDIAT, and BLUI datasets demonstrate the superior performance of FET-UNet compared to state-of-the-art methods. The model achieves Dice coefficients of 82.9% on BUSI, 88.9% on UDIAT, and 90.1% on BLUI.
Conclusion: FET-UNet shows great potential to advance breast ultrasound image segmentation and support more precise clinical diagnoses. Further research could explore the application of this framework to other medical imaging modalities and its integration into clinical workflows.

1. Introduction

Breast cancer poses a significant health risk to women around the world, with its high incidence and mortality rates highlighting the urgent need for advancements in early detection and precise diagnostic methods [1]. Breast ultrasound imaging is a widely used diagnostic tool due to its non-invasive nature, cost-effectiveness, and safety, particularly in differentiating benign from malignant lesions [2]. However, manual segmentation of breast ultrasound images is labor-intensive, subject to inter-observer variability, and often lacks consistency. Therefore, developing automated, accurate, and robust segmentation methods is essential for enhancing diagnostic accuracy and improving patient outcomes.

Traditional image segmentation techniques in breast ultrasound imaging, such as region-growing [3,4], thresholding [5,6], and clustering [7], have significant limitations. These methods heavily depend on hand-crafted features and are highly susceptible to noise and artifacts prevalent in ultrasound images, leading to suboptimal performance and inconsistent results. With the advent of deep learning, CNNs have become the cornerstone of image segmentation due to their ability to learn hierarchical features directly from data, outperforming traditional methods. UNet [8] and its variants are among the most successful CNN-based architectures for medical image segmentation. They employ an encoder-decoder structure with skip connections, allowing for precise localization and segmentation. Additionally, other UNet-like architectures such as UNet++ [9], U2Net [10], and SA-UNet [11] have emerged due to their simplicity and outstanding performance. Despite their success, CNN-based methods primarily focus on local feature extraction and often struggle to capture long-range dependencies and global context, which are crucial for accurately segmenting complex structures in breast ultrasound images. To address these limitations, some studies have attempted to enhance the network's ability to capture global

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context by expanding the receptive field [12–14] or incorporating attention mechanisms [15,16] to focus on important features.

Originally designed for natural language processing, transformers have recently shown significant promise in revolutionizing computer vision tasks. Vision Transformers [17] and Swin Transformers [18] capture global context through self-attention mechanisms, providing a powerful means of modeling long-range dependencies. Although transformers can effectively capture global contextual information, they often neglect local feature details. Furthermore, transformer-based methods tend to require large-scale datasets to achieve satisfactory results. Recently, researchers have investigated the integration of CNNs with transformer architectures for medical image segmentation. One prevalent method involves incorporating transformer blocks into the feature extraction phase, substituting high-level convolutional modules, as exemplified by techniques such as TransUNet [19] and LevitUNet [20]. However, this approach can lead to suboptimal fusion of CNN and transformer modules, as they operate separately at low and high levels respectively, lacking information interaction across multiple scales. This limitation hampers the extraction of rich detailed features, global contextual information, and accurate identification of lesions that closely resemble surrounding tissues.

To address these challenges, we propose a novel hybrid network, fusion-enhanced transformer UNet (FET-UNet), specifically tailored for breast ultrasound image segmentation. Our approach leverages the localized feature extraction capabilities of CNNs and the global contextual modeling strengths of transformers to enhance segmentation performance. The FET-UNet architecture consists of parallel branches for feature extraction and downsampling: one utilizing ResNet [21] blocks and the other employing Swin Transformer blocks. To synergistically combine the features from both branches, we introduce a novel advanced feature aggregation module (AFAM). This module effectively integrates the features from the ResNet and Swin Transformer branches across both spatial and channel dimensions, resulting in a comprehensive and robust feature representation. The decoder, designed with Residual Blocks and a multi-scale upsampling mechanism, incorporates skip connections to ensure detailed and accurate segmentation outputs.

The main contributions of this paper are as follows:

- We propose a hybrid network that integrates CNNs and Swin Transformers within a UNet-like framework to enhance breast ultrasound image segmentation.
- We introduce a novel advanced feature aggregation module (AFAM) to combine the features from ResNet and Swin Transformer branches, resulting in a comprehensive feature representation.
- We demonstrate the effectiveness of FET-UNet through extensive experiments on breast ultrasound image segmentation benchmarks, showing that it outperforms existing state-of-the-art methods.

2. Related work

2.1. CNNs-based breast lesion segmentation methods

Recently, the swift advancement of deep learning has given rise to numerous CNN-based approaches for breast tumor segmentation. Yap et al. [22] were pioneers in employing CNN techniques for breast lesion detection, utilizing a modified Fully Convolutional Network (FCN) for segmentation, which demonstrated superiority over traditional methods. To further enhance segmentation accuracy, Hu et al. [12] proposed the use of dilated convolutions in deeper networks to expand the receptive field in breast tumor segmentation. Vakanski et al. [26] made an early attempt to add attention mechanism in CNNs in the breast tumor segmentation task. Through the attention module, the network features can be spatially ranked according to the size of attention, thus improving the attention of the network to high attention areas and inhibiting their attention in less significant areas. Xue et al. [23] introduced the concept of boundary detection and designed a global guidance block to improve the accuracy of breast lesion segmentation by learning long range information. Huang et al. [24] approached lesion segmentation in breast ultrasound images from a boundary-rendering perspective. Their framework includes a boundary selection module that focuses on edge areas and a GCN-based boundary rendering module that converts nodule boundaries into graph data. Lou et al. [25] addressed the semantic gap issue in U-shaped CNNs by introducing adaptive multi-scale feature extraction and contextual correlation techniques. They improved feature interactions between encoder and decoder stages, leading to refined feature fusion and enhanced multi-level contextual information. AAU-net [15] developed a hybrid attention module that adaptively adjusts spatial and channel attention, allowing the network to autonomously learn the weight parameters for these two types of attention. This enhances the network's ability to extract features across these different dimensions. Qi et al. [41] proposed the Multi-scale Dynamic Fusion Network (MDF-Net) for breast ultrasound tumor segmentation, using a two-stage architecture with multiscale feature selection and refinement to improve segmentation accuracy and robustness.

Based on CNNs-based methods, significant efforts have been made to improve breast tumor segmentation accuracy by incorporating multi-scale feature extraction, expanding the receptive field with dilated convolutions, and integrating attention mechanisms to enhance focus on relevant areas. However, CNNs inherently struggle with capturing long-range dependencies and contextual information, and face challenges in handling complex variations in tumor appearance and shape.

2.2. Transformers in medical image segmentation

The evolution of transformers in computer vision has led researchers to increasingly apply this technology to medical image processing. TransFuse [27] integrates transformers and CNNs with a late fusion approach, leveraging CNNs for spatial correlation and transformers for global relationship modeling. UNETR [28] constructs a simple 3D medical image segmentation network, reconstruct the input 3D images to 1D sequence and extracts features using a stack of transformers as the encoder. Swin-Unet [29] constructs a U-shaped architecture dedicated to medical image segmentation, leveraging Swin Transformer blocks as the foundational building elements. Valanarasu et al. [30] introduced MedT, which incorporates a gated axial attention layer for multi-head attention in medical image segmentation, along with a Local-Global (LoGo) training strategy to capture global context from full images and finer details from patches. HiFormer [31] bridges CNNs and transformers in medical image segmentation by combining global information from the Swin Transformer and local details from a CNN encoder, facilitated by a Double-Level Fusion (DLF) module for enhanced feature integration. HAU-Net [33] presents a hybrid framework that incorporates a CNN-transformer architecture, embedding a Local-Global transformer block within the skip connections of a U-shaped network. Additionally, they added a cross attention block on the decoder side, facilitating interaction across different layers. CMFF-Net [34] also adopts a hybrid approach, proposing a cross-attention feature fusion module that merges the global context information from transformers with the local spatial information from CNNs. They also designed a feature extraction module to refine local features. Zhou et al. [42] introduced a full-resolution residual stream and a deep supervision mechanism into TransU-Net for breast ultrasound image segmentation. The residual stream preserves full-resolution features across different levels to enhance feature fusion, while deep supervision mitigates gradient dispersion. Additionally, the transformer module suppresses irrelevant features and improves feature extraction. EH-Former [43] introduces a region-wise Curriculum Learning approach to dynamically adjust focus on hard regional features in breast ultrasound (BUS) images. The model employs a two-stage structure with uncertainty estimation for dividing regional difficulty, an Adaptive Easy-Hard region Separator (AdaSep) for separating features of varying difficulties, and a

Dynamic Easy-Hard Feature Fusion (D-Fusion) module to progressively fuse features based on training epochs.

These examples illustrate the trend of integrating transformers with CNN architectures to harness their combined strengths, improving both local feature extraction and global context modeling. Building on these advancements, our proposed FET-UNet aims to enhance breast ultrasound image segmentation by merging the localized feature extraction of CNNs with the global contextual capabilities of transformers. By introducing the AFAM, FET-UNet achieves superior segmentation performance, advancing the current state-of-the-art in hybrid models.

3. Method

In this work, we propose a novel hybrid network named FET-UNet, designed to integrate CNN and Swin Transformer architectures within a UNet-like framework for improved image segmentation. FET-UNet leverages the complementary strengths of convolutional networks for capturing local features and transformers for modeling global context. This architecture consists of parallel branches for feature extraction and downsampling: one utilizing ResNet blocks and the other employing Swin Transformer blocks. A novel advanced feature aggregation module (AFAM) is introduced to synergistically combine these features, resulting in a comprehensive and robust feature representation. The decoder is designed with Residual Blocks and features a multi-scale upsampling mechanism with skip connections to ensure detailed and accurate segmentation outputs.

3.1. Overview

The overall architecture of FET-UNet is illustrated in Fig. 1. The network follows a symmetric encoder-decoder design, typical of UNet architectures. This network integrates CNNs and Swin Transformer architectures within a UNet-like framework to enhance image segmentation tasks. FET-UNet leverages the localized feature extraction capabilities of CNNs and the global contextual modeling strengths of transformers. The architecture consists of parallel branches for feature extraction and downsampling, followed by a sophisticated fusion mechanism that synergistically combines these features. The decoder is designed with residual blocks and a multi-scale upsampling mechanism that incorporates skip connections, ensuring detailed and accurate segmentation results. In the following, we detail each component of the proposed FET-UNet.

3.2. Encoder

The encoder of FET-UNet comprises two parallel pathways: the ResNet branch and the Swin Transformer branch. Each branch processes the input image to extract features and perform downsampling, which are then combined using a novel fusion mechanism.

3.2.1. ResNet branch

The ResNet branch begins with a convolutional stem, followed by four stages of residual layers. Each stage consists of a residual block followed by average pooling for downsampling. Residual blocks are essential for capturing and refining local features efficiently while maintaining a manageable computational complexity. This design ensures that detailed local features are preserved and propagated through the network.

3.2.2. Transformer branch

Parallel to the ResNet branch, the Swin Transformer branch starts with a patch partitioning and linear embedding layer, followed by four layers of Swin Transformer blocks. Each transformer layer is composed of several Swin Transformer blocks, designed to capture long-range dependencies and perform patch merging to downsample the features.

Each Swin Transformer block, as illustrated in Fig. 2(a), consists of

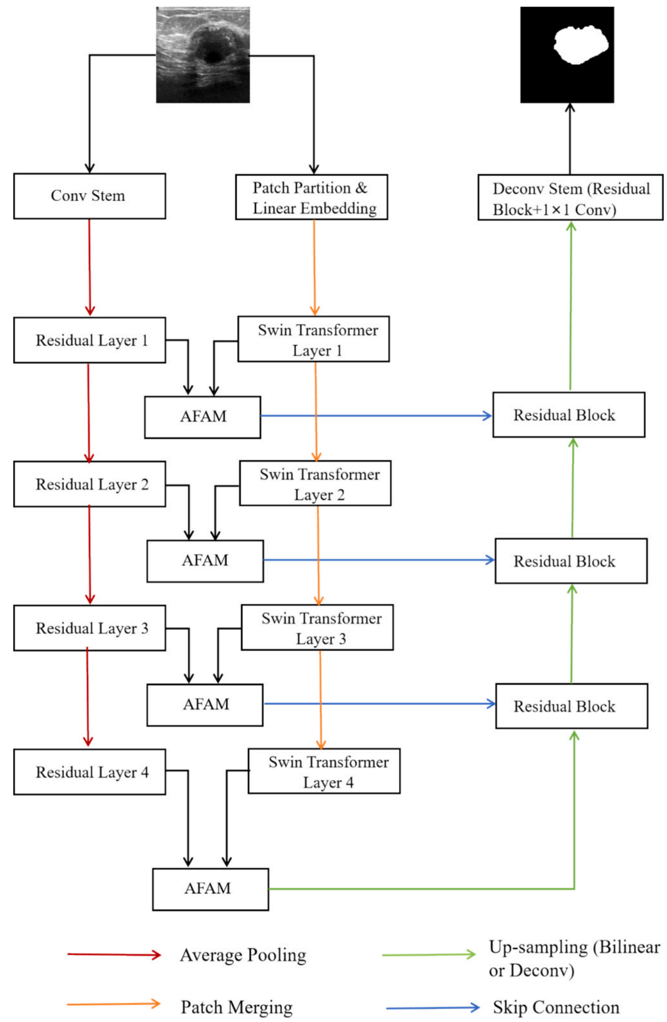

Fig. 1. The architecture of the proposed FET-UNet which integrates CNN and Swin Transformer branches, using AFAM modules to effectively fuse features. The decoder mirrors the encoder structure with Residual Blocks and upsampling, connected by skip connections to maintain spatial detail.

two main components. The first component is the window-based multi-Head self-attention (W-MSA) module. This module performs self-attention within non-overlapping local windows, effectively capturing local dependencies and reducing computational complexity. The second component is the shifted window-based multi-head self-attention (SW-MSA) module, which shifts the windows from the previous step and applies self-attention again. This shifting mechanism allows for information exchange between neighboring windows, thereby enabling the network to model long-range dependencies and achieve global attention.

Specifically, within each Swin Transformer block, the W-MSA is applied first to compute self-attention within fixed-size windows. This is followed by layer normalization (LN) and a multi-layer perceptron (MLP) to refine the features. Next, the windows are shifted, and the SW-MSA is applied to enable cross-window interactions. Another round of LN and MLP is then applied to further refine the features. Mathematically, the process within a Swin Transformer block can be described as follows:

$$\hat{z}^l = W - MSA(LN(z^{l-1})) + z^{l-1} \tag{1}$$

$$z^l = MLP\left(LN\left(\hat{z}^l\right)\right) + \hat{z}^l \tag{2}$$

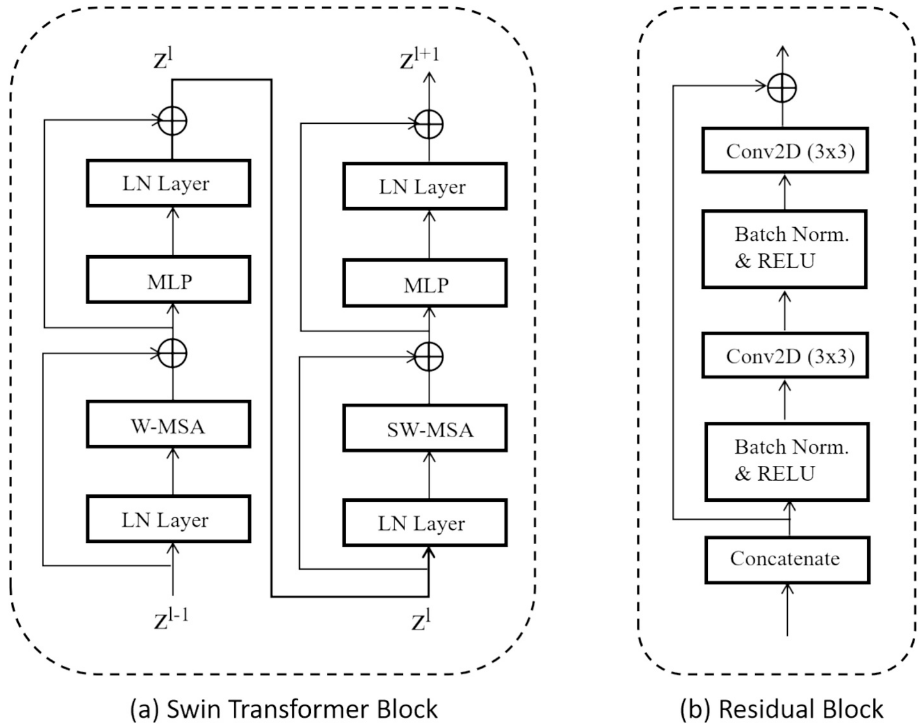

Fig. 2. (a) Structure of the Swin Transformer Layer in the encoder. (b) Structure of the Residual Block in the decoder.

$$\hat{z}^{l+1} = \mathrm{SW-MSA}\left(\mathrm{LN}\left(z^l\right)\right) + z^l \tag{3}$$

$$z^{l+1} = MLP\left(LN\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1} \tag{4}$$

3.2.3. Advanced feature aggregation module (AFAM)

To effectively combine the features extracted by the ResNet and Swin Transformer branches, we introduce the AFAM at each corresponding stage of the encoder. The AFAM is designed to merge features from both branches, leveraging their complementary strengths to enhance feature representation and improve segmentation accuracy. The structure of this module is illustrated in Fig. 3.

The fusion process starts with the concatenation of features from the corresponding stages of the ResNet and Swin Transformer branches. Given a ResNet feature $F^{res} \in R^{C\times H\times W}$ and a Swin Transformer feature $F^{swin} \in R^{C\times HW}$, the Swin Transformer feature is first reshaped to $R^{C\times H\times W}$, and then concatenated with the ResNet feature. This concatenation ensures that both the detailed local features captured by the ResNet branch and the global contextual information captured by the Swin Transformer branch are retained in the merged feature map. Finally, a 1 $\times$ 1 convolution is applied to the concatenated features to reduce the dimensionality back to C, resulting in the combined feature F^c.

$$F^c = Conv\left(Concat\left(F^{res}, F^{swin}\right)\right) \tag{5}$$

The combined feature is then processed by a multi-head self-attention (MSA) module. Specifically, the combined feature is first processed with a 1 $\times$ 1 convolution to reduce its dimensionality to 1 $\times$ H $\times$ W. Following this step, it passes through a MSA module to compute self-attention, aiming to learn the spatial weight distribution of the combined feature by capturing long-range dependencies. In the MSA module, the feature is mapped to three different representations: the query (Q), key (K), and value (V) vectors. These are obtained through linear projections of the input feature. The self-attention is then calculated by computing the dot product between Q and K to measure the similarity, followed by a softmax operation to obtain the attention weights. These weights are applied to the V vectors through element-wise multiplication, producing the attended feature. Next, the obtained spatial attention weight map is used to reweight the combined feature F^c through element-wise multiplication, and a residual connection is applied to obtain the spatial attention-weighted fused feature $F^{c-atten}$. This process can be represented as follows:

$$F^{c-atten} = \mathrm{MSA}(\mathrm{Conv}(F^c)) \times F^c + F^c \tag{6}$$

Subsequently, the spatial attention-weighted fused feature is passed through a Squeeze-and-Excitation (SE) [35] block for channel attention

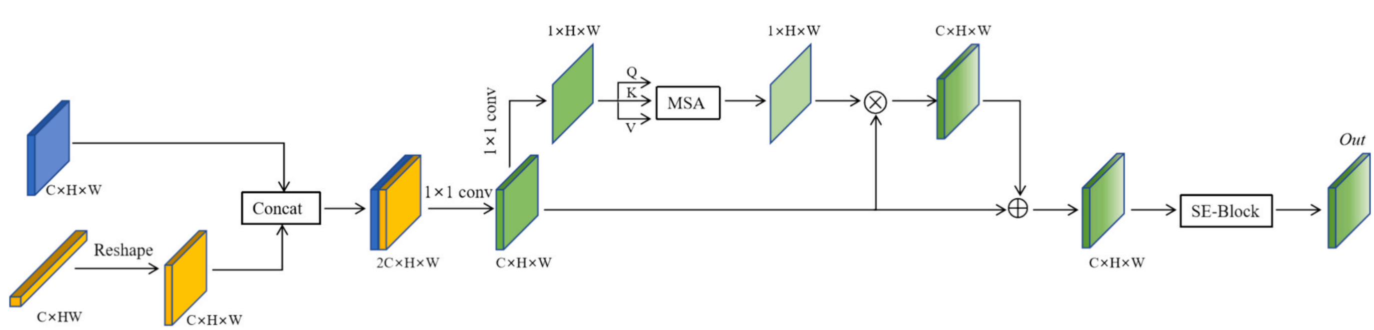

Fig. 3. The structure of our proposed Advanced Feature Aggregation Module (AFAM).

weighting, which enhances the most informative channels while suppressing less relevant ones. This process ensures that the final fused feature map is both robust and informative, leading to the final output of the AFAM.

$$F^{afam} = SE(F^{c-atten}) \quad (7)$$

The AFAM integrates MSA for spatial attention and SE for channel-wise recalibration, ensuring that both spatial and channel-wise important features are emphasized. This comprehensive approach to feature fusion significantly enhances the network's ability to perform accurate image segmentation by leveraging the strengths of both the ResNet and Swin Transformer branches.

3.3. Decoder

In each layer of the decoder, we upsample the fused features produced by the AFAM and concatenate them with the features from the previous layer's AFAM output. This concatenated feature is then refined using a Residual Block, as depicted in Fig. 2(b). Since the AFAM features already integrate characteristics from both the ResNet and Swin Transformer layers in the encoder, we use a simplified residual structure in the Residual Block to enhance computational efficiency and processing speed. Our decoder incorporates three residual block layers and a deconv stem. In the final deconv stem layer, the ultimate segmentation prediction is obtained through a residual block followed by a 1 × 1 convolution.

The proposed FET-UNet effectively combines the localized feature extraction capabilities of CNNs with the global contextual understanding of transformers, offering a robust solution for image segmentation tasks. The integration of the novel AFAM further enhances feature representation, leading to improved segmentation accuracy. This hybrid architecture is expected to perform well across various image segmentation benchmarks, demonstrating the effectiveness of merging convolutional and transformer paradigms.

4. Experiments and results

4.1. Datasets

This research assessed the suggested technique using three publicly accessible breast ultrasound datasets: BUSI [36], UDIAT [37], and BLUI [38]. Each dataset includes breast ultrasound images with annotated lesion information, providing a valuable resource for training and evaluation. The datasets are summarized as follows:

The BUSI dataset, collected in 2018 at Baheya Hospital, Cairo, Egypt, contains 780 breast ultrasound images from 600 female patients aged 25 to 75. The images were captured using the LOGIQ E9 and LOGIQ E9 Agile ultrasound systems, with a resolution of 1280 × 1024 pixels, and are stored in PNG format. The dataset is categorized into three classes: benign (437 images), malignant (210 images), and normal (133 images). Ground truth annotations were provided by radiologists through freehand segmentation in Matlab, with segmentation masks saved using the "_mask" suffix in the filename. For the purpose of this study, we focused on the benign and malignant images for training and validation.

The UDIAT dataset, collected in 2012 from the UDIAT Diagnostic Centre of the Parc Taulí Corporation in Sabadell, Spain, consists of 163 images from different women, each potentially containing one or more lesions. The images were obtained using a Siemens ACUSON Sequoia C512 system with a 17L5 HD linear array transducer (8.5 MHz). The dataset includes both malignant and benign lesions, with 53 images containing malignant masses and 110 images containing benign lesions. Specifically, the malignant lesions include 40 invasive ductal carcinomas, 4 ductal carcinomas in situ, 2 invasive lobular carcinomas, and 7 unspecified malignant lesions, while the benign lesions consist of 65 unspecified cysts, 39 fibroadenomas, and 6 other benign lesions. Ground truth annotations were provided through lesion delineations made by experienced radiologists, ensuring the accuracy of the lesion labels. The original image size is 760 × 570 pixels.

The BLUI dataset released by Shahid Beheshti University of Medical Sciences in Tehran includes images acquired from consecutive patients who presented to a cancer imaging center. It includes 123 and 109 ultrasound images of malignant and benign breast lesions, all from women aged 18 and older who had no history of breast cancer in either breast. Each image contains one or more lesions, and the ground truth annotations were provided by an experienced radiologist through freehand segmentation. The original image sizes vary, and preprocessing was applied to resize and standardize the images for consistency.

To ensure reliable and consistent evaluation, a five-fold cross-validation was implemented for all three datasets. Each dataset was split into training and testing sets using a 4:1 ratio. Images from the same patient were consistently assigned to either the training or test set within each fold to avoid bias and prevent data leakage. Comparisons were made using identical splitting methods across all datasets. The exact number of images used in the train and test sets for each dataset is summarized in the Table 1.

In addition, the distribution of tumor region sizes is also crucial for training and testing. To provide a clearer understanding, we summarize the tumor region size distribution for each dataset in Fig. 4. The tumors in each dataset are classified into three categories based on the proportion of the tumor region relative to the total image area: tumors occupying less than 5 %, between 5 % and 20 %, and greater than 20 % of the image area. To maintain representativeness, we employed stratified splitting to preserve the tumor size distribution (as shown in Fig. 4) between training and test sets across all folds. This ensures that the proportions of tumors occupying less than 5 %, 5 %-20 %, and greater than 20 % of the image area remain consistent in both sets, preventing bias due to uneven distribution.

4.2. Implementation details

In our study, we adopt ResNet34 and Tiny Swin Transformer as the backbone networks. The ResNet34 architecture starts with a 7 × 7 convolutional layer, followed by four residual stages composed of 3, 4, 6, and 3 residual blocks, respectively. Each block includes convolutional layers, batch normalization, and ReLU activation, with the output channels for the stages being 256, 512, 1024, and 2048, respectively. For the Tiny Swin Transformer, the network begins with patch partitioning and linear embedding, followed by four stages of Swin Transformer blocks, with 2, 2, 6, and 2 blocks in each stage. The output channels of these stages are 96, 192, 384, and 768, and the number of attention heads in each stage is 3, 6, 12, and 24, respectively.

We conducted all experiments within the PyTorch framework, training them on a single NVIDIA GeForce RTX 4090 GPU equipped with 24 GB of graphics memory. Our approach utilized the Adam optimizer with a learning rate that started at 0.0001 and gradually decreased to a minimum of 0.00001 during training. The models were trained for 100 epochs with a batch size of 16. To standardize input dimensions, all images were resized to 224 × 224 pixels. These hyperparameters were determined through systematic preliminary experiments aimed at maximizing segmentation performance while ensuring training stability and convergence. This process with the selected parameters demonstrated an optimal balance between accuracy and computational

Table 1
Dataset splits for training and validation.

Dataset	Total (images)	Train (images)	Test (images)	Splitting Method
BUSI	647	518	129	4:1 Train-Test Split
UDIAT	163	130	33	
BLUI	232	186	46	

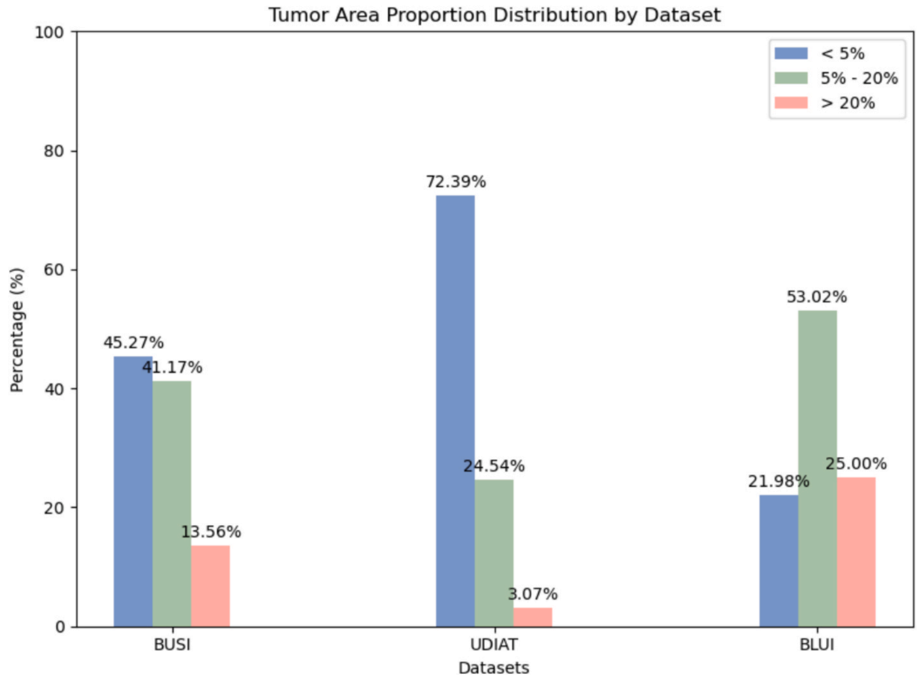

Fig. 4. Tumor region size distribution across datasets, categorized by relative area: less than 5%, 5%-20%, and greater than 20%.

efficiency.

Given the typically small size of medical image datasets, we implemented various data augmentation strategies to prevent overfitting and enhance the models' generalization ability. Data augmentation was dynamically applied during each training iteration, generating diverse image variations to enhance model generalization. These strategies included random rotations between 0 and 10 degrees, horizontal flips, vertical flips, and center cropping. Fig. 5 illustrates examples of augmented images, showcasing the variety introduced by our augmentation techniques. For our study, we used weights pre-trained on ImageNet to initialize the parameters of the ResNet34 and tiny Swin Transformer modules. It is important to note that, except for UNet, all other comparative methods used CNN or transformer backbones with pre-trained weights of the same level as those in our approach.

4.3. Loss function

In tackling the significant class imbalance inherent in medical image segmentation, we implemented a novel hybrid loss function. This function seamlessly blends binary cross-entropy (BCE) loss and Dice loss, harnessing their complementary attributes to enhance gradient descent stability and improve segmentation accuracy. Specifically, the hybrid loss function is expressed mathematically as follows:

$$L_{BCE} = -\frac{1}{N}\sum_{i=1}^{N}[y_i log(p_i) + (1-y_i)log(1-p_i)] \tag{8}$$

$$L_{Dice} = 1 - \frac{2\sum_{i=1}^{N} y_i p_i}{\sum_{i=1}^{N}(y_i + p_i)} \tag{9}$$

$$L = \frac{L_{BCE} + L_{Dice}}{2} \tag{10}$$

where N represents the total pixel count of the input image. $y_i \in \{0, 1\}$ signifies the ground-truth label assigned to the i-th pixel. The predicted probability $p_i \in [0, 1]$ signifies the likelihood that the i-th pixel belongs to the positive class. By combining the BCE and Dice losses, this hybrid loss function benefits from the stable training provided by BCE and the effective handling of class imbalance by Dice loss. This ensures that the segmentation model is both robust and capable of achieving high accuracy, effectively managing the difficulties posed by unbalanced class distributions.

4.4. Evaluation metrics

To affirm superiority and structural validity, we applied six widely recognized segmentation metrics to evaluate the accuracy of segmentation across all breast ultrasound datasets. These metrics include the

Fig. 5. Examples of augmented images.

Dice Similarity Coefficient (Dice), 95 % Hausdorff Distance (Hd), Intersection over Union (IoU), Accuracy (Acc), Specificity (Spe), and Precision (Pre), providing a comprehensive framework for evaluation. These metrics can be formulated as follows:

$$Dice = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{11}$$

$$HD = max(h(pred, gt), h(gt, pred)) \tag{12}$$

$$IoU = \frac{TP}{TP + FP + FN} \tag{13}$$

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{14}$$

$$Spe = \frac{TN}{TN + FP} \tag{15}$$

$$Pre = \frac{TP}{TP + FP} \tag{16}$$

where TP represents the count of correctly segmented breast lesion pixels and TN represents the count of correctly segmented background pixels. FP indicates pixels incorrectly segmented as background instead of breast lesion pixels, while FN denotes breast lesion pixels incorrectly predicted as background. The function $h(A, B)$ is defined as $max_{a \in A}\{min_{b \in B}\|a - b\|\}$, where $\| \bullet \|$ indicates the euclidean distance between two pixels.

4.5. Evaluation results

To comprehensively evaluate the superiority of the proposed method, we conducted two types of experiments. The first type involved comparative experiments on individual datasets, specifically comparing our method with nine state-of-the-art segmentation methods on the BUSI, UDIAT, and BLUI datasets. The second type focused on generalization ability. Since the three datasets used in this study are all breast ultrasound datasets, we merged these three datasets into a single dataset and performed evaluation experiments on this combined dataset. Additionally, to further validate the network's performance on tumors with different characteristics, we divided the BUSI dataset into benign and malignant subsets and conducted comparative experiments. To ensure fairness and reliability, all experiments were performed using five-fold cross-validation, and all methods used the same experimental and parameter settings. Furthermore, to assess statistical significance, we performed paired t-tests comparing our method to the second-best method on Dice and Hd95 metrics across folds, considering differences significant when the p-value was less than 0.05.

4.5.1. Comparison with state-of-the-art methods

Tables 2, 3, and 4 present the comparative results of different methods on the BUSI, UDIAT, and BLUI datasets, respectively. Our FET-UNet outperforms other competing methods across all evaluation metrics, except for the ACC metric on the BLUI dataset where it ranks second. Among the nine methods compared, UNet [8], R34-DeepLabv3 [39], MCRNet [25], nnU-Net [44], and AAU-Net [15] are entirely CNN-based, SwinUnet [30] is based on the transformer architecture, and TransUNet [19], FAT-Net [40], and ATFE-Net [32] utilize hybrid architectures combining CNNs and transformers.

MCRNet shows the best performance among the CNN-based methods, likely due to its use of multi-scale dilated convolutions which capture multi-scale contextual information and enhance the relationships between the encoder and decoder, thereby improving segmentation accuracy. TransUNet and ATFE-Net, both hybrid methods that combine CNNs and transformers, also perform well by enhancing the extraction of global information through the addition of transformer modules to the higher layers of the CNN encoder. SwinUnet, on the other hand, performs relatively poorly across all three datasets. This may be because it relies entirely on Swin Transformer modules for both the encoder and decoder without using CNNs, which can miss local details that are crucial for smaller datasets like breast ultrasound images, thereby affecting segmentation performance.

FET-UNet uses parallel branches of CNNs and transformers to extract image features, and the proposed AFAM block effectively integrates these features, capturing long-range dependencies more effectively. Compared to the second-best method MCRNet, FET-UNet improves the Dice coefficient by 0.93 % and reduces the HD by 2.63 mm on the BUSI dataset. On the UDIAT dataset, FET-UNet outperforms ATFE-Net by 1.29 % in Dice coefficient and reduces the HD by 1.43 mm compared to FAT-Net. On the BLUI dataset, compared to the second-best method TransUNet, FET-UNet improves the Dice coefficient by 1.22 % and reduces the HD by 0.70 mm. Regarding statistical significance, FET-UNet demonstrates p-values below 0.05 for both Dice ($p = 0.03$) and Hd95 ($p = 0.02$) on the UDIAT dataset, confirming a robust advantage over the second-best method. On the BUSI dataset, the Dice p-value of 0.07 suggests performance akin to MCRNet, whereas the Hd95 p-value of 0.01 indicates a substantial improvement. Similarly, on the BLUI dataset, FET-UNet achieves a significant edge with a Dice p-value of 0.04 and an Hd95 p-value of 0.01 relative to TransUNet. Taken together, these findings underscore FET-UNet's reliable performance across three public ultrasound datasets, with a pronounced strength in boundary precision validated by statistical analysis.

Fig. 6 illustrates the visual segmentation outcomes obtained by various methods across three breast ultrasound image datasets. Taking the first and second rows of images as examples, most methods struggle to distinguish the tumor from surrounding tissues due to the similar grayscale distribution, leading to noticeable false positives. Only ATFE-Net and our method accurately locate the target in the first row, while in the second row, our method is the only one that achieves accurate segmentation. The fourth and fifth rows contain images where the target volumes are relatively small, leading to missed detections and false positives in many methods. Our method and FatNet accurately detect the target in the fourth row, while in the fifth row, only our method and

Table 2
Evaluation results (mean ± std) of different methods on the BUSI dataset.

Method	Dice	Hd95	IoU	Acc	Spe	Pre
UNet	70.7 ± 6.1	29.4 ± 4.9	61.1 ± 3.9	94.9 ± 0.9	97.3 ± 0.6	75.2 ± 3.3
R34 DeepLabv3	80.9 ± 3.2	13.8 ± 1.9	73.1 ± 2.6	96.2 ± 0.6	97.7 ± 0.5	81.9 ± 3.7
MCRNet	82.2 ± 2.7	12.4 ± 1.2	74.1 ± 2.1	96.7 ± 0.5	97.8 ± 0.4	82.6 ± 3.0
nnU-Net	81.1 ± 3.1	14.8 ± 5.1	73.7 ± 3.9	96.6 ± 0.9	97.8 ± 0.7	82.2 ± 3.5
AAU-Net	78.5 ± 2.3	15.8 ± 3.1	71.3 ± 2.0	95.8 ± 0.5	97.5 ± 0.4	80.5 ± 2.9
TransUNet	81.0 ± 2.8	14.6 ± 4.2	73.3 ± 2.8	96..2 ± 0.8	97.5 ± 0.8	81.8 ± 2.4
SwinUnet	79.7 ± 3.3	15.4 ± 3.7	70.4 ± 3.6	96.0 ± 0.4	97.3 ± 0.4	80.1 ± 3.0
FAT-Net	81.7 ± 2.6	13.0 ± 3.1	73.3 ± 3.3	96.4 ± 0.5	97.6 ± 0.5	81.9 ± 3.2
ATFE-Net	82.1 ± 3.0	12.8 ± 2.0	73.8 ± 3.4	96.6 ± 0.7	97.6 ± 0.5	81.7 ± 3.8
FET-UNet	**82.9 ± 2.3**	**9.8 ± 2.0**	**74.7 ± 2.4**	**96.8 ± 0.2**	**98.6 ± 0.4**	**87.1 ± 3.5**
P-values	0.07(Dice), 0.01(Hd95)					

Table 3
Evaluation results (mean ± std) of different methods on the UDIAT dataset.

Method	Dice	Hd95	IoU	Acc	Spe	Pre
UNet	81.1 ± 4.4	12.4 ± 3.8	72.9 ± 6.4	98.3 ± 0.7	99.2 ± 0.5	78.3 ± 5.9
R34 DeepLabv3	87.4 ± 3.3	5.4 ± 3.0	78.9 ± 3.3	98.8 ± 0.5	99.4 ± 0.3	85.8 ± 3.4
MCRNet	87.6 ± 2.8	5.7 ± 2.6	79.6 ± 2.6	98.8 ± 0.5	99.4 ± 0.4	85.9 ± 2.4
nnU-Net	86.1 ± 3.6	5.4 ± 4.1	78.5 ± 5.3	98.8 ± 0.8	99.5 ± 0.5	85.7 ± 4.2
AAU-Net	83.5 ± 2.5	8.7 ± 3.0	75.6 ± 3.3	98.5 ± 0.4	99.4 ± 0.6	83.3 ± 3.3
TransUNet	87.5 ± 2.0	5.9 ± 3.3	79.8 ± 1.7	98.8 ± 0.4	99.4 ± 0.3	86.1 ± 2.3
SwinUnet	85.9 ± 1.8	6.3 ± 4.3	76.2 ± 2.8	98.5 ± 0.7	99.3 ± 0.5	84.4 ± 3.5
FAT-Net	87.6 ± 1.8	5.2 ± 4.9	79.4 ± 4.0	98.9 ± 0.6	99.4 ± 0.4	85.6 ± 3.3
ATFE-Net	87.8 ± 1.4	5.6 ± 2.8	79.8 ± 3.5	98.9 ± 0.5	99.6 ± 0.3	86.4 ± 3.1
FET-UNet	**88.9 ± 2.0**	**3.8 ± 3.2**	**81.6 ± 2.8**	**99.1 ± 0.4**	**99.7 ± 0.2**	**88.5 ± 3.0**
P-values	0.03(Dice), 0.02(Hd95)					

Table 4
Evaluation results (mean ± std) of different methods on the BLUI dataset.

Method	Dice	Hd95	IoU	Acc	Spe	Pre
UNet	85.0 ± 3.1	9.8 ± 3.2	76.9 ± 4.0	95.5 ± 0.7	97.6 ± 0.9	85.1 ± 1.9
R34 DeepLabv3	87.8 ± 1.8	6.4 ± 2.1	80.7 ± 1.7	96.4 ± 0.2	97.7 ± 0.7	87.8 ± 2.0
MCRNet	88.1 ± 2.5	6.3 ± 3.2	80.6 ± 1.3	96.8 ± 0.4	97.7 ± 0.5	86.3 ± 1.1
nnU-Net	88.2 ± 2.2	6.3 ± 2.0	80.9 ± 2.1	96.7 ± 0.6	97.6 ± 0.3	88.1 ± 1.3
AAU-Net	86.9 ± 3.1	7.3 ± 3.6	78.7 ± 3.3	96.2 ± 0.7	97.4 ± 0.7	86.0 ± 2.1
TransUNet	89.1 ± 1.6	5.7 ± 1.9	82.0 ± 1.6	**97.0 ± 0.2**	97.6 ± 0.5	88.3 ± 2.6
SwinUnet	87.4 ± 1.5	6.7 ± 2.7	80.2 ± 1.9	96.6 ± 0.5	97.3 ± 0.6	87.6 ± 1.9
FAT-Net	88.4 ± 1.9	6.2 ± 1.3	81.0 ± 3.8	96.8 ± 0.5	97.7 ± 0.6	87.4 ± 1.6
ATFE-Net	88.3 ± 1.8	6.5 ± 3.2	81.2 ± 2.6	96.7 ± 0.6	97.6 ± 0.8	88.1 ± 3.0
FET-UNet	**90.1 ± 1.4**	**5.0 ± 1.2**	**82.8 ± 1.3**	96.9 ± 0.4	**98.3 ± 0.4**	**90.1 ± 1.6**
P-values	0.04(Dice), 0.01(Hd95)					

ATFE-Net do so, with our method providing more precise segmentation results. When dealing with irregular target contours, as shown in the sixth, seventh, and eighth rows, our method excels in capturing details and achieves the most accurate segmentation results. The visual segmentation results demonstrate that our proposed method outperforms others when facing challenges such as similar grayscale distributions, irregular targets, and small target volumes.

4.5.2. Generalization performance verification

To verify the generalization ability of our proposed method, we combined the BUSI, UDIAT, and BLUI ultrasound datasets into a single dataset and conducted comparative experiments on this merged dataset. To ensure a balanced distribution of tumor types (benign and malignant) across the training and testing sets, we applied stratified cross-validation during the dataset splitting. This stratification ensured that each fold contained a representative proportion of benign and malignant tumor cases. Table 5 presents the results of these experiments, showing that our method outperformed all others across six metrics. Specifically, our method improved the Dice coefficient by 0.86 % and reduced the HD by 1.73 mm compared to ATFE-Net, the second-best method.

Furthermore, to further evaluate the model's performance in different scenarios, we split the BUSI dataset into benign and malignant tumor subsets. Table 6 shows the comparison results for these subsets. Malignant tumors, with their irregular shapes and rough edges, pose more significant segmentation challenges. As a result, all methods performed better on the benign subset than on the malignant one. However, our method showed superior performance on both subsets, with a marked improvement in malignant tumor segmentation. In this case, our method increased the Dice coefficient by 1.49 % and decreased the HD by 2.91 mm compared to the next best method.

In terms of statistical significance, the merged dataset reveals a Dice p-value of 0.09 indicating performance similar to ATFE-Net, yet an Hd95 p-value of 0.03 confirms a notable improvement. For benign tumors, a Dice p-value of 0.08 suggests equivalence to ATFE-Net, while an Hd95 p-value of 0.04 denotes a significant gain. For malignant tumors, FET-UNet exhibits distinct superiority with p-values below 0.05 for both Dice ($p = 0.02$) and Hd95 ($p = 0.01$) over the next-best methods. Overall, these results affirm FET-UNet's consistent generalization and enhanced capability in diverse and complex segmentation tasks as supported by statistical evidence.

4.6. Ablation study

Table 7 presents the results of the ablation experiments conducted on the BUSI dataset using our proposed method. In these experiments, we kept the decoder unchanged and first used either ResNet34 or Swin Transformer alone as the encoder. Next, we used both ResNet34 and Swin Transformer in the encoder but fused them with a simple concatenation. Finally, we applied our proposed AFAM.

As shown in Table 7, using Swin Transformer alone as the encoder resulted in poor performance, whereas using ResNet34 alone yielded significantly better results. This indicates that for small medical image datasets, Transformers alone cannot effectively leverage their strength in capturing long-range dependencies and, in doing so, miss out on the CNN's ability to extract detailed features. When we used concatenation to fuse ResNet34 and Swin Transformer, the segmentation results showed only a slight improvement compared to using ResNet34 alone. However, when we applied our proposed AFAM, the network performance improved significantly. The Dice coefficient increased by 2.32 %, and the HD decreased by 4.8 mm compared to using concatenation. This demonstrates the effectiveness of AFAM, which leverages spatial Multi-Head Self-Attention (MSA) and channel Squeeze-and-Excitation (SE) to effectively and fully integrate features from both CNN and transformer, thereby enhancing the overall network performance.

5. Discussion

The FET-UNet represents a significant step forward in the integration of CNN and Transformer architectures for medical image segmentation. The design philosophy behind FET-UNet is to harness the local feature extraction strengths of CNNs and the global context modeling capabilities of Transformers. This dual approach is especially crucial in breast

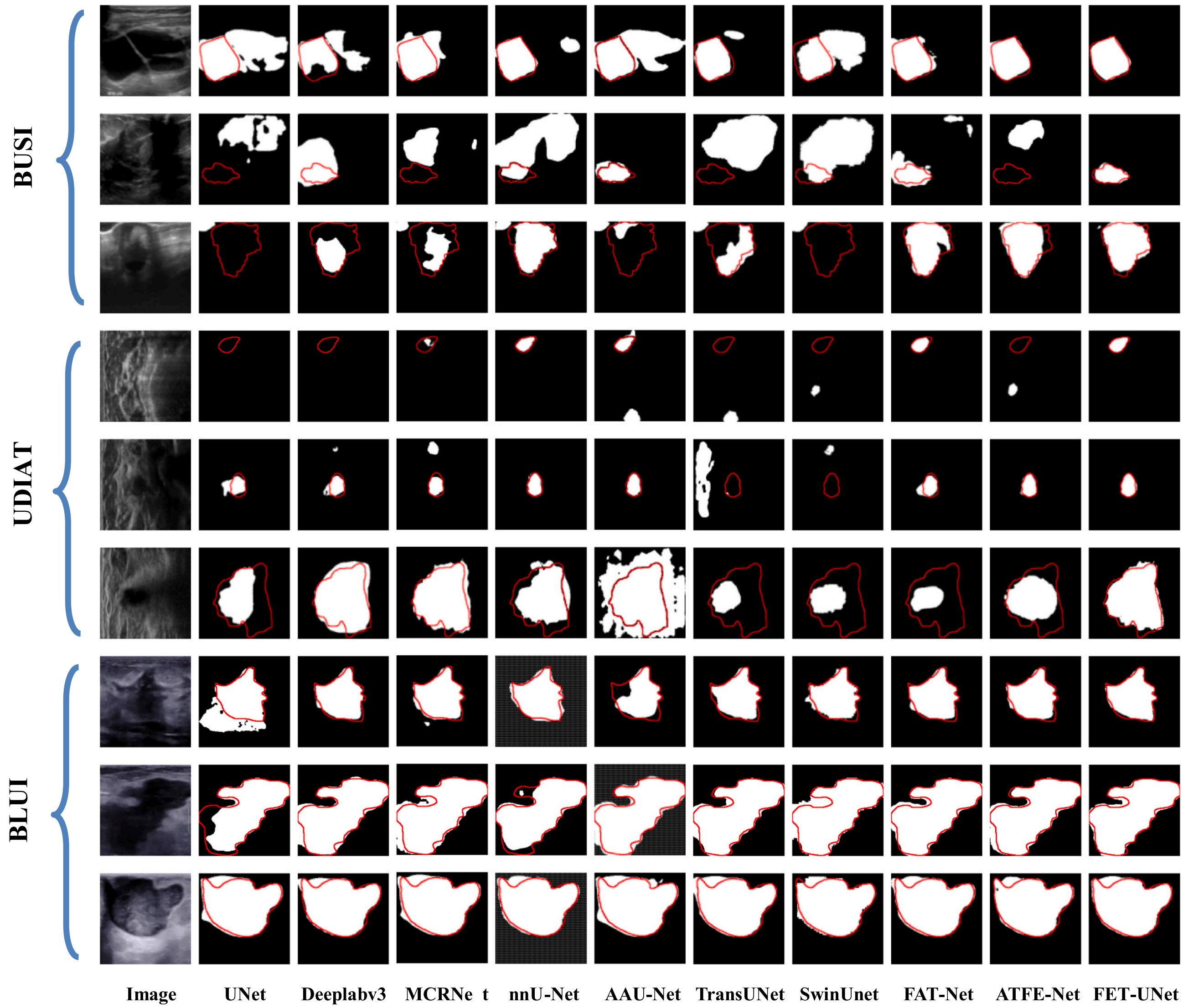

Fig. 6. Visualization of segmentation results using various methods on three ultrasound datasets. The red curve outlines the ground truth. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 5
Performance comparison (mean ± std) of different method on the merged dataset.

Method	Dice	Hd95	IoU	Acc	Spe	Pre
UNet	74.2 ± 8.1	20.3 ± 6.2	66.0 ± 7.7	96.8 ± 0.7	98.3 ± 0.4	75.1 ± 9.7
R34 DeepLabv3	81.8 ± 2.9	12.6 ± 3.1	74.2 ± 3.7	97.1 ± 0.4	98.2 ± 0.6	81.6 ± 4.3
MCRNet	82.4 ± 2.6	12.0 ± 2.5	74.7 ± 3.2	97.1 ± 0.6	98.4 ± 0.4	81.9 ± 3.9
nnU-Net	80.2 ± 5.2	12.8 ± 4.2	74.1 ± 3.7	97.2 ± 0.7	98.3 ± 0.6	80.9 ± 4.7
AAU-Net	77.9 ± 4.4	14.9 ± 4.2	69.3 ± 4.1	96.9 ± 0.7	98.2 ± 0.5	78.0 ± 5.2
TransUNet	82.0 ± 3.1	12.4 ± 2.7	74.4 ± 3.8	96.6 ± 0.8	98.2 ± 0.3	81.7 ± 4.3
SwinUnet	79.7 ± 4.8	13.5 ± 4.8	71.7 ± 4.3	97.0 ± 0.7	98.1 ± 0.4	80.9 ± 5.4
FAT-Net	81.8 ± 4.2	12.7 ± 2.0	74.3 ± 3.0	97.2 ± 0.8	98.5 ± 0.3	81.9 ± 4.2
ATFE-Net	82.5 ± 3.7	11.3 ± 2.5	75.0 ± 3.0	97.4 ± 0.5	98.6 ± 0.4	82.7 ± 3.5
FET-UNet	**83.2 ± 3.4**	**9.6 ± 2.2**	**75.7 ± 2.8**	**97.6 ± 0.5**	**98.6 ± 0.4**	**84.1 ± 4.3**
P-values	0.09(Dice), 0.03(Hd95)					

ultrasound image segmentation, where precise delineation of tumor boundaries is critical for accurate diagnosis and treatment planning.

Our method introduces the AFAM, which plays a pivotal role in merging features from ResNet34 and Swin Transformer branches. This module not only preserves the distinct advantages of both architectures but also facilitates effective interaction between local and global features. The effectiveness of AFAM is underscored by the substantial performance gains observed in our experiments. One of the key insights from our work is the recognition of the limitations inherent in using either CNNs or Transformers in isolation. While CNNs excel at capturing fine-grained details, they often fall short in modeling long-range dependencies. Conversely, Transformers are adept at understanding global context but may overlook intricate local features, especially in smaller datasets. FET-UNet's hybrid architecture successfully bridges this gap,

Table 6
Performance comparison (mean ± std) for benign and malignant lesions in the BUSI dataset across different methods.

Method	Benign		Malignant	
	Dice	Hd95	Dice	Hd95
UNet	76.8 ± 2.0	18.5 ± 2.4	61.6 ± 4.3	31.6 ± 8.2
DeepLabv3	83.2 ± 2.7	11.0 ± 2.7	75.4 ± 3.1	22.2 ± 7.7
MCRNet	82.8 ± 2.4	12.6 ± 3.8	76.5 ± 4.3	20.3 ± 6.6
nnU-Net	83.2 ± 2.0	11.7 ± 2.8	75.2 ± 4.4	23.4 ± 7.9
AAU-Net	80.3 ± 3.0	14.9 ± 3.7	74.1 ± 4.2	25.7 ± 7.4
TransUNet	83.2 ± 1.9	11.5 ± 2.0	75.6 ± 3.9	22.1 ± 8.2
SwinUnet	82.2 ± 2.9	14.5 ± 3.2	75.1 ± 4.7	23.1 ± 4.9
FAT-Net	83.3 ± 2.6	12.0 ± 2.1	76.4 ± 3.8	21.6 ± 5.9
ATFE-Net	83.4 ± 1.8	9.7 ± 3.1	76.4 ± 5.2	22.0 ± 6.8
FET-UNet	**84.2 ± 1.9**	**8.7 ± 2.0**	**77.6 ± 2.3**	**17.4 ± 6.6**
P-values	0.08(Dice), 0.04(Hd95)		0.02(Dice), 0.01(Hd95)	

Table 7
Ablation study on different encoder configurations.

Encoder	Dice	Hd95	IoU	Acc	Spe	Pre
Swin Transformer	69.6 ± 4.7	25.1 ± 3.8	59.7 ± 2.7	95.0 ± 0.7	97.3 ± 1.0	71.9 ± 3.5
ResNet34	80.7 ± 2.6	14.8 ± 2.1	72.6 ± 3.0	96.1 ± 0.7	97.1 ± 0.8	80.8 ± 2.1
Fused by Concatenation	81.1 ± 2.1	14.6 ± 2.7	72.7 ± 2.4	95.7 ± 0.6	97.1 ± 0.6	82.1 ± 2.7
Fused by AFAM	**82.9 ± 2.3**	**9.8 ± 2.0**	**74.7 ± 2.4**	**96.8 ± 0.2**	**98.6 ± 0.4**	**87.1 ± 3.5**
P-values	0.02(Dice), 0.01(Hd95)					

providing a balanced approach that enhances segmentation accuracy.

The generalization experiments conducted on merged and tumor-specific datasets further highlight the robustness of FET-UNet. The ability to maintain high performance across diverse datasets and tumor characteristics suggests that FET-UNet is not only versatile but also resilient to variations in imaging conditions and tumor morphology. This robustness is crucial for clinical applications, where variability is a constant challenge.

Despite its advantages, FET-UNet is not without limitations. One of the primary challenges is the increased computational complexity and memory usage due to the dual-branch architecture and the AFAM module. This complexity can make the model less suitable for real-time applications or deployment in resource-constrained environments. Future work could focus on optimizing the network to reduce its computational footprint without compromising performance. Techniques such as model pruning, quantization, or the development of more efficient attention mechanisms could be explored to achieve this. Furthermore, while the AFAM module effectively merges CNN and Transformer features, there is room for improvement in the feature fusion strategy. Advanced fusion techniques that dynamically weigh the importance of features from each branch or incorporate more complex interactions between features could further enhance the network's performance. Additionally, FET-UNet currently relies on fully supervised learning, which requires large amounts of annotated data. In clinical settings, obtaining such data can be challenging and time-consuming. Future work could explore semi-supervised or unsupervised learning approaches to reduce the dependency on annotated data.

6. Conclusion

In this paper, we propose FET-UNet which offers a novel and effective solution for breast ultrasound image segmentation by integrating the complementary strengths of CNNs and transformers. The innovative AFAM is key to this integration, enabling the network to capture both local details and global context with high precision. The extensive experiments conducted across multiple datasets confirm that FET-UNet outperforms existing state-of-the-art methods, demonstrating superior segmentation performance and generalization capabilities. The network's robustness in handling both benign and malignant tumors, particularly in complex cases, highlights its potential for enhancing clinical diagnostic accuracy. FET-UNet represents a significant advancement in medical image segmentation, providing a powerful tool that can improve diagnostic workflows and patient outcomes. Future work will focus on further optimizing the network architecture and exploring its application to other medical imaging tasks, thereby extending its impact and utility in the field of medical diagnostics.

Data availability statement

The data presented within this article are available from the corresponding author on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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